Patient Record Similarity Retrieval

Technical Report

1. Project Overview

The objective of this project is to develop and evaluate a text-based retrieval system that identifies patient records with similar clinical descriptions. The system uses TF-IDF as the baseline representation and compares its retrieval results with a second information-retrieval method, BM25. For each patient record, both methods rank the other records in the collection and return the five highest-ranked matches.

2. Dataset Description

The supplied **PMC_Patients** collection contains **8,579 patient records**. All records in the supplied collection were processed; no subset sampling was used. After text preprocessing, the average document length was **322.17 tokens** and the median was **289 tokens**. The shortest processed record contained 19 tokens, while the longest contained 2,028 tokens. Length was calculated by splitting each processed text on whitespace and counting the resulting tokens.

3. Text Preprocessing

Clinical text was standardized before vectorization and retrieval. The preprocessing pipeline performs the following operations:

- **Lowercasing:** converts text to lowercase so that capitalization differences do not create separate terms.
- **Punctuation removal:** removes punctuation that is not useful for the selected term-based representation.
- **Tokenization:** separates the text into individual word tokens.
- **Stopword removal:** removes common English words that generally contribute little to distinguishing records.
- **Lemmatization:** applies NLTK WordNet lemmatization to reduce words to their base forms where supported.

Negation terms such as *no* are retained because they can change the meaning of clinical statements. For example, a statement indicating that a patient has no evidence of a condition should not be treated as equivalent to a positive statement about that condition.

Packages and tools: Python; NLTK for English stopwords and WordNet lemmatization; scikit-learn for TF-IDF vectorization and cosine similarity; NumPy and SciPy for numerical and sparse-matrix operations; and Python standard-library modules for JSON processing, regular expressions, file handling, and BM25 index construction.

4. Task 1: TF-IDF Representation

TF-IDF (term frequency-inverse document frequency) assigns a weight to each term based on its frequency within a patient record and its distribution across the full collection. A term receives greater importance when it is frequent in a particular record but less common across the dataset. The resulting sparse matrix has **8,579 document rows** and **30,473 term features**.

Parameter	Setting	Purpose and rationale
smooth_idf	True	Smooths inverse document-frequency values and avoids unstable weighting for terms with low document frequency.
sublinear_tf	True	Applies logarithmic scaling to term frequency so repeated occurrences have a diminishing effect.
norm	L2	Normalizes each document vector, making cosine similarity suitable for comparing vector direction.
min_df	2	Excludes terms appearing in only one document, reducing the influence of extremely rare terms.
max_df	0.95	Excludes terms appearing in more than 95% of documents because they provide limited distinguishing information.
max_features	50,000	Places an upper limit on vocabulary size to control memory use and processing cost. The resulting vocabulary contained 30,473 terms.
dtype	float32	Uses 32-bit floating-point values to reduce matrix memory requirements.

The matrix is saved in SciPy sparse **.npz** format so it can be loaded directly into Python without converting the full matrix to a dense representation. The accompanying metadata file stores the document and feature mappings needed to associate matrix rows with patient UIDs and columns with terms.

5. Task 2: Additional Retrieval Method

BM25 Ranking

BM25 is a term-based ranking method that scores a document according to the query terms it contains. Its score considers three main factors: how informative a term is across the collection, how frequently the term appears in a document, and the document's length. Unlike a simple count of matching words, BM25 limits the additional benefit of repeatedly using the same term and adjusts for differences in document length.

The implementation builds an **inverted index**. For each term, the index stores the document IDs in which that term occurs and the term's contribution to the document score. This allows a query to accumulate scores from documents containing its terms without comparing every term against every document independently.

BM25 settings: k1 = 1.5 controls term-frequency saturation. A higher term frequency can increase a document's score, but the benefit gradually levels off. b = 0.75 controls document-length normalization; it balances the effect of a document's length against the average document length. The implementation uses a positive inverse-document-frequency expression so that terms occurring in many documents receive lower weights.

For each patient, its own preprocessed tokens are used as the query against the indexed collection. The query-term iteration uses a set, so repeated query terms are counted once in scoring. This is a simplified treatment of query-term frequency and is a detail of the implemented BM25-style retrieval method.

6. Similarity and Ranking Strategy

TF-IDF: cosine similarity is used to compare the normalized TF-IDF vectors. Cosine similarity measures the angle between two vectors and focuses on the pattern of term weights rather than simply favoring longer records. It is appropriate for normalized document vectors, but it mainly captures lexical overlap. Two clinically related records that use different wording may therefore receive a low similarity score.

BM25: documents are ranked by their BM25 scores. A higher score indicates a stronger match to the query terms under the BM25 scoring formula. BM25 accounts for term frequency and document length, but it is also term-based and does not independently understand clinical meaning, synonyms, or semantic equivalence.

Both methods rank all other patient records and return the five highest-scoring records. The query patient's own record is explicitly excluded from its results.

7. Task 3: Evaluation Method

The *similar_patients* field supplied with each record is used as the reference set of relevant patients. Relevance values 1 (moderate similarity) and 2 (high similarity) are treated equally as relevant, as specified in the assignment.

- **Precision@5:** the number of relevant records among the five retrieved records, divided by 5. It measures how many of the returned recommendations are relevant.
- **Recall@5:** the number of relevant records retrieved in the top five, divided by the total number of relevant records listed for that patient. It measures how much of the known relevant set was retrieved.
- **Averaging:** Precision@5 is averaged across all 8,579 patients. Patients with no listed similar patients are excluded from the Recall@5 average, and their number is reported.
- **Top-five overlap:** the number of patient UIDs shared by the TF-IDF and BM25 top-five lists, divided by 5. This indicates how similarly the two methods rank their recommendations.

The evaluation values are calculated separately for every patient and retrieval method. The per-patient results are then saved in JSON, with aggregate averages saved as a summary.

8. Results

The complete pipeline was executed on all 8,579 patient records. The mean Precision@5 and Recall@5 results for the two retrieval methods are shown below.

Evaluation measure	TF-IDF	BM25
Mean Precision@5	0.0078 (0.78%)	0.0062 (0.62%)
Mean Recall@5	0.0336 (3.36%)	0.0276 (2.76%)
Patients included in recall average	4,802	4,802
Patients excluded from recall average	3,777	3,777

Mean top-five overlap: 55.92%. On average, approximately 2.8 of the five recommendations were shared between the two methods. This indicates that the methods produced some similar recommendations, while also differing in a substantial portion of their ranked results.

9. Interpretation of Results

TF-IDF achieved higher average scores than BM25 in this experiment. Its mean Precision@5 was 0.78%, compared with 0.62% for BM25. Its mean Recall@5 was 3.36%, compared with 2.76% for BM25. In practical terms, the top-five recommendations from either method contained relatively few records that were explicitly identified as relevant in the provided reference data. The average recall values also indicate that the top-five lists captured only a small proportion of each patient's listed relevant records.

These results show that TF-IDF performed better than BM25 under the selected preprocessing, scoring settings, dataset, and evaluation procedure. They do not establish that TF-IDF is universally superior. Different representations, clinical terminology handling, or semantic retrieval methods could produce different results.

10. Ground-Truth Coverage and Limitations

A separate coverage check identified **11,490** ground-truth patient references across the records. Of these, only **540 (4.70%)** referred to UIDs present in the supplied 8,579-record collection; the remaining **10,950** references were not present in the collection. Consequently, many reference patients were unavailable as retrieval candidates. This restricts which labeled matches the system could return and is an important limitation when interpreting the measured precision and recall.

The Recall@5 denominator was kept as the total number of similar patients listed in each source record, rather than being reduced to only references found in the collection. This follows the assignment's definition of recall. The 3,777 patients with no listed similar patients were excluded from the mean recall calculation and retained in the precision calculation.

11. Output Files and Validation

- **Task 1:** *tfidf_matrix.npz* stores the TF-IDF matrix; *tfidf_metadata.json* stores the associated row and feature mappings.
- **Task 2:** *patient_similarities.json* stores each patient's top-five recommendations from TF-IDF and BM25.
- **Task 3:** *evaluation_results.json* stores per-patient Precision@5 and Recall@5 values; *evaluation_summary.json* stores aggregate results.

Validation covered all 8,579 records. Each record had five recommendations for both methods, and no patient was recommended as its own match. The validation reported zero errors. The project can be executed from its root directory with **python main.py data/PMC_Patients**.

Conclusion. This project implements a modular patient-record retrieval pipeline using TF-IDF with cosine similarity and BM25 ranking. TF-IDF achieved the stronger evaluation scores in this experiment. The limited coverage of ground-truth references within the available dataset is a key constraint on the measured results.